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Introduction

Background & Statement of Problem

Background

Water is the most valuable natural resource and plays a vital role in mainly maintaining a healthy ecosystem, supporting biodiversity. Freshwater bodies such as **rivers, lakes, and canals** are home to numerous aquatic organisms. However, these water bodies are increasingly threatened by pollution, climate change, and excessive nutrient inputs, which can lead to environmental problems such as harmful algal growth.

Algae are naturally occurring photosynthetic organisms found in freshwater and marine ecosystems. Under normal conditions, they are an essential part of aquatic ecosystems because they produce oxygen through photosynthesis and form the base of many aquatic food chains. However, when Nutirents such as **Nitrogen** and **Phosphorus** become excessively available due to agricultural **runoff, wastewater discharge, urban stormwater** or other human activities, algae can multiply rapidly. Combined with warm temperatures, sufficient sunlight, and slow-moving water, these conditions often result in excessive algal growth, commonly known as **algal blooms**.

Excessive algae growth has many negative **environmental, social and economic consequences**. Thick ayers of algae reduce the penetration of sunlight into the water, preventing submerged aquatic plants from carrying out photosynthesis. As algae die and decompose, microorganisms consume large amounts of dissolved oxygen, leading to oxygen with depletion that can stress or kill fish and other aquatic organisms. In severe cases, algal blooms may produce **unpleasant odours, reduce water clarity, clog waterways, interfere with recreational activities**, and significantly decrease the aesthetic value of natural environments.

The **Spoy Canal in Kleve, Germany**, Germany, experiences recurring algae growth, making it an important environmental challenge for the city. Besides being part of hte local drainage and water management system, the canal is an attractive feature of residents and visitors and contributes to the city's landscape. Because the canal passes through public areas, maintaining its **ecological health and visual appearance is essential**

As part of this initiative, **Hochschule Rhein-Waal (HSRW)** collaborates with the city of kleve by involving engineering students in solving real-world sustainability challenges. Instead of simply proposing theoretical ideas, students apply the **VDI2221 Engineering design methodology** as well as the **VDI2220 Engineering design methodology**, which emphasizes systematic product developement. The design process begins by understanding the *problem, analysing stakeholders, identifying customer needs*, and translating these needs into measurable engineering requirements before generateing design concepts.

In this project, students are expected to develop a **sustainable engineering solution** capable of reducing algae accumulation in the **Spoy Canal** while balancing technical *performance, environmental protection, economic feasibility, safety, maintainability, and sustainability*.

Statement of Problem

The Spoy Canal currently experiences recurring algae accumulation, which negatively affects the ecological conditions and visual quality of the waterway. Excessive algae reduce water quality, decrease dissolved oxygen concentrations, interfere with aquatic ecosystems, and create an unpleaseant environment for both residents and visitors. The Problem also increases maintenance efforts and cost of the local authorities responsible for managing the canal.

As Kleve prepares for **LaGa 2029**, the existing algae problem presents additional challenges. A canal covered with algae is inconsistent with the objectives of **LaGa**, which seeks to promote **environmental sustainability, biodiversity and attractive urban landscape**. Therefore, an effective and sustainable solution is required to improve the condition of the canal before and during the event.

Traditional algae removal methods often require consdierable **manual labour, repeated maintenance, and operational costs**. Some methods may also disturb aquatic habitats or consume significant amounts of energy. Consequently, there is a need for an innovative engineering solution that removes algae efficiently while minimizing environmental impacts.

The proposed solution should not only remove algae effectively but also operate safely in public areas. Consume little energy, protect aquatic organisms, resist corrosion in freshwater environments, require minimal maintenance, and comply with environmental and safety regulations. Since the machine will potentially operate in areas visited by the public during **LaGa 2029**, it should also have an **aesthetically acceptable appearance and produce minimal noise**.

Following the engineering design methodology, the first stage of the project is not to select a specific but to define what the product must achieve. This involves identifying stakeholder needs and translating them into measurable engineering requirements that will guide concept generation and evaluation in the following stages of the project.

Requirements and Stakeholders

The overall engineering challenge is therefore to design a sustainable, reliable, safe, and efficient algae-removal machine capable of improving the ecological condition of visual quality of the **Spoey Canal** while satisfying technical, environmental, economic, and stakeholder requirements. Such a solution would contribute to the long-term management of the canal and support the environmental objectives of **LaGa 2029**.

Identifying the different stakeholders, their interests, their relationship to the project, and their respective priorities and categories is essential for effective project planning and management.

[Figure 1](#) presents the stakeholders considered in the project and provides an overview of their respective interests, estimated project impact, and priority.

Based on the identified stakeholders, a set of project requirements is established. These requirements relate to both the characteristics of the final product and the project development process. A weight is assigned to each requirements based on the degree of importance it is for the project and the total sum of all weights must equal **100%**. 2 revisions of this table exist. The final revision identifies the requirements seen on [Figure 2](#).

The differences between these revisions are pointed out below:

1. Additions (Revision 2)

- **Itemized Legal Framework (ID 17-19)** Replaced the generic local compliance entry with three distinct legal permits: Federal Waterway Permit (Strom- und Schifffahrtspolizeiliche Genehmigung, 4% weight), Water Use Permit (Wasserrechtliche Erlaubnis, 4% weight), and Biodiversity & Habitat Protection (Naturschutzrecht, 3% weight).
- **Canal Navigation Preservation (ID 22):** Added a safety requirement mandating that operation must not disrupt waterway navigation, evaluated via allowable traffic stoppage time (Demand, 3% weight).
- **Infrastructure Protection (ID 23)** Introduced an ergonomics requirement barring any physical alteration, reconstruction, or destruction of existing canal walls or bank infrastructure (Demand, 5% weight).
- **Weather Resistance (ID 24)** Added a performance requirement demanding continuous system operability under hot, sunny, moist, and rainy conditions during peak algal bloom periods (Demand, 2% weight).

2. Changes & Modifications (Revision 1 → Revision 2)

Technical & Operational Scale-up - algae Removal Rate (ID 1): Target extraction rate increased from 20–30 kg/h to 120–200 kg/h.

- **Storage Capacity (ID 7):** Onboard holding capacity expanded from 40 kg to 200–250 kg.
- **Machine Mass (ID 8):** Maximum mass threshold increased from < 80 kg to < 500 kg.
- **Surface Footprint (ID 12):** Maximum envelope area expanded from 1 m² to < 5 m².
- **Operating Speed (ID 3):** Reduced from ~5 km/h to ~1 km/h to optimize harvesting efficiency.
- **Energy Limit (ID 5):** Refined from nominal 500 W to an explicit ceiling of < 500 W.

Interest and Impact Table				
Primary (P) or Secondary (S)	STAKEHOLDER	NEEDS / INTERESTS	Estimates Project Impact	Estimated priority
Primary	Owner (HSRW)	1. Reputation 2. Innovative sustainable solutions 3. Advertisement	Med + High + Low +	3
Primary	Team members (Students)	1. Satisfaction of accomplishment 2. New set of skills 3. Possible future opportunities(enlargement of the project)	Low + High + Med +	6
Primary	Citizens	1. Clean environment 2. Sound pollution (avoid) 3. Good looking product	High + Med - Med +	3
Primary	City Municipality	1. Sustainable solution to pond algae Problem	High +	3
Primary	Sponsor(city)	1. Promoting sustainable technologies 2. Positive publicity of the city	High + Med +	5
Secondary	LaGa 2029	1. Safe product 2. Visually appealing product 3. Minimal disruption during operation	High + Med + Med +	7
Secondary	Tourists	1. Pleasant recreational environment 2. Environmentally friendly Technology	High + High +	6
Secondary	University Community	1. Educational value 2. Reliable solution (to algae problem) 3. Visually appealing product.	High + High + Low +	7
Secondary	Biodiversity	1. Sustainable pond Ecosystem 2. Preservation of Natural habitat 3. Minimal disturbance from the product	High + High + Med +	8
Secondary	Businesses	1. Minimal disruption during business operations	Med +	2
Secondary	Insurance company	1. Safe and reliable product 2. Low liability risk 3. Compliance with safety regulations	High + High + High +	9

Figure 1: Stakeholder List

Week 1 Deliverable

Requirements List Project / Product							
	Project:	AlgaNova					
	Group:	9					
	Revision:	2					
	Date:	7/24/2026					
ID	Requirement	Category	Target / Criterion	Unit	Demand / Wish	Weight	Comments
1	Algae removal	Function	120-> 200	Kg/h	D	8%	Removal efficiency
2	Water depth operation	Geometry	0 - 1	m	D	7%	Range of operation
3	Operating speed	Performance	~1	Km/h	D	5%	Ensures efficient cleaning
4	Noise emission	Ergonomics	~60	dB	W	3%	Low noise during operation.
5	Energy consumption	Sustainability	<500	W	D	6%	Energy-efficient operation.
6	Operating Duration	Performance	4->6	h	D	4%	Continuous operation for 4 hours.
7	Algae storage capacity	Function	200 -> 250	Kg	D	5%	Reduces unloading frequency.
8	Machine weight	Geometry	<500	Kg	W	7%	Lightweight and portable.
9	Regular Maintenance (Cleaning)	Cost	100	Euros	D	4%	Easy and low-cost maintenance.
10	Harmful emissions	Safety	None	None	D	5%	Environmentally friendly operation.
11	Corrosion Resistance	Manufacturing	Visual Inspection	None	W	3%	Rust-resistant materials used.
12	Compact design (Surface Area)	Geometry	<5	m^2	W	4%	Easy to transport and store.
13	Attractive appearance	Ergonomics	public opinion on design	None	W	3%	Modern and user-friendly design
14	Durability	Other	10	years/warranty	D	3%	Long service life.
15	Machine Safety & Electrical and Radio Regulations	Other	Class 1	None	D	3%	Meets Class 1 standards.
16	Battery Safety Regulations	Other	UN 38.3 Certification.	None	D	3%	Complies with UN 38.3
17	Comply to the Federal Waterway Permit (Strom- und Schifffahrtpolizeiliche Genehmigung)	Other	Conform to legal requirements	None	D	4%	
18	Water Use Permit (Wasserrechtliche Erlaubnis)	Other	Conform to legal requirements	None	D	4%	
19	Biodiversity & Habitat Protection (Naturschutzrecht)	Other	Conform to legal requirements	None	D	3%	
20	Scalability	Manufacturing	Adaptable to Canal width	m	W	2%	Adaptable to different canal widths.
21	Water proof	Safety	IPX6	None	D	4%	IPX6 water-resistant design.
22	Must not block canal Navigation	Geometry	Stoppage time of other simultaneously occurring activities	min	D	3%	Simultaneous operation of other canal activities
23	Preserve canal infrastructure	Ergonomics	No reconstruction nor destruction of the current canal's infrastructure	None	D	5%	Not aiming at altering the canals infrastructure to accommodate the system
24	weather resistant	Performance	Should Operable in hot,sunny,moist and rainy weather conditions	None	D	2%	The system will be mostly operational during warm sunny weathers, as we experience a heavy algal infestations during these periods
25					Total Weight Check	100%	
26						OK	

Figure 2: Requirements List

1. Design

1. Functional Design

Core Principles

Functional Design defines **what** a system must do before establishing physical solutions or technical mechanisms.

- **Solution Neutrality:** Functions are defined without committing prematurely to specific technologies or physical components.
- **System Boundaries & Flows:** The overall function maps system inputs into outputs across three primary categories:
 - **Energy (E):** Electrical, mechanical, or thermal energy transfers.
 - **Signal (S):** Control inputs, sensor data, or status feedback.
 - **Mass / Matter (M):** Materials, fluids, or workpieces passing through the system.
- **Function Categories:** Core operational actions used to structure sub-functions:
 - *Conversion* (changing input form to output form)
 - *Varying* (amplifying, stepping down, or modulating)
 - *Connecting/Disconnecting* (switching, coupling, or mixing)
 - *Channeling* (directing or transferring power, signals, or mass)
 - *Storing* (holding energy, signals, or mass)

Below is the Functional design diagram showcasing the different functions, sub-functions, signals, Matter(mass), and Energy flowing into the system.

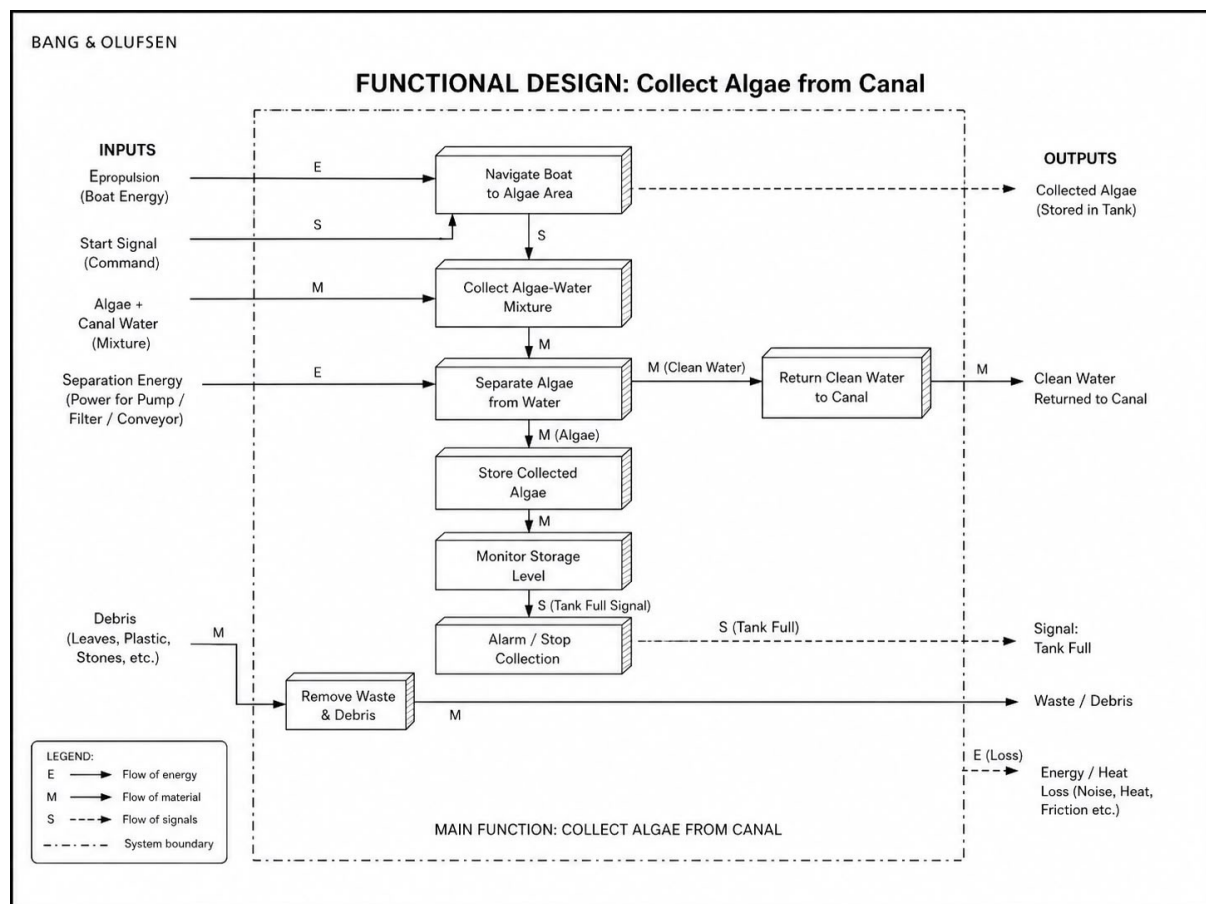


Figure 3: Functional Design Revision 1

The diagram above shows the initial design of system itself. After some more analysis, we came up with a better design which ended up being our final design of the system shown seen on the figure below.

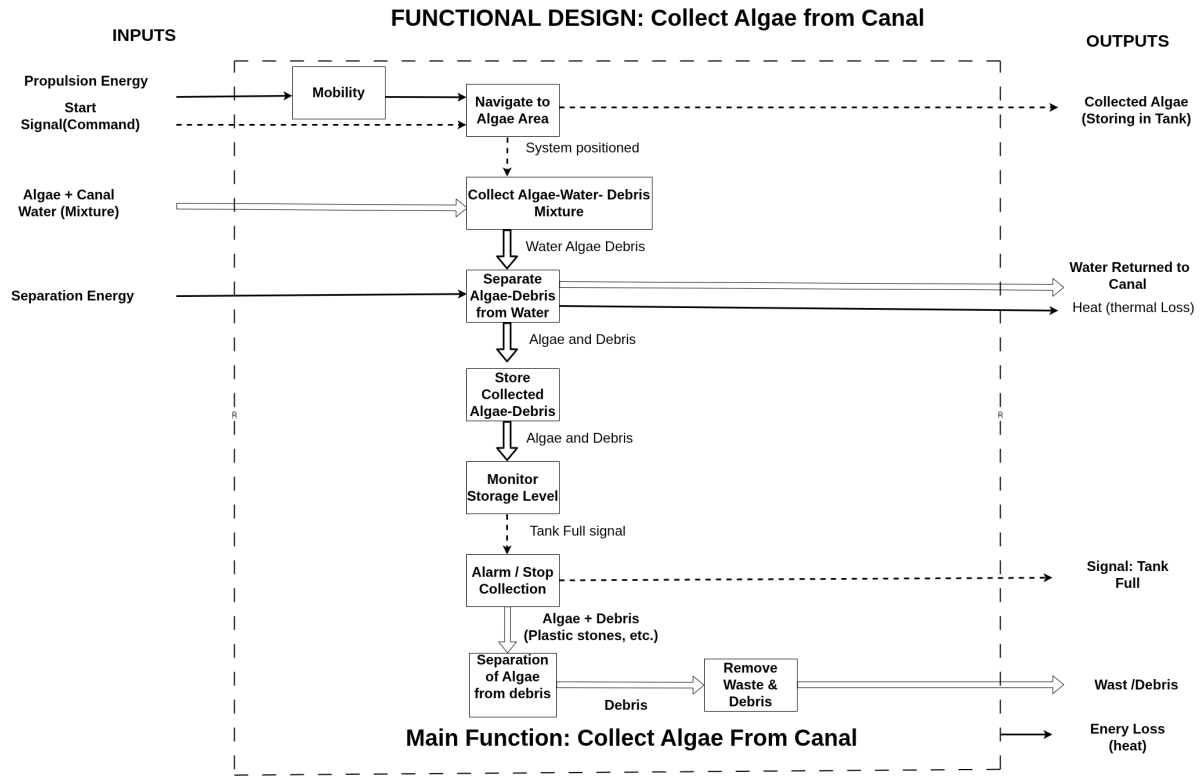


Figure 4: Functional Design Revision 2

Below the functional design diagram is a table explaining the various changes and modifications done to generate the final functional diagram.

Feature	Initial Diagram	Current Diagram	Reason for Change
Navigation	Not clearly defined	Navigate to Algae Area	Ensures the machine reaches the correct cleaning location before starting the cleaning process.
System Positioning	Not included	System Positioned	Confirms that the machine is correctly positioned before algae collection begins.
Collection Process	General algae collection	Collect Algae-Water-Debris Mixture	Represents the actual operating condition where algae, water, and floating debris are collected together.
Separation Process	General separation process	Separate Algae-Debris from Water	Makes the separation process more specific and realistic by distinguishing water from solid materials.
Water Output	Not clearly represented	Water Returned to Canal	Indicates that filtered water is discharged back into the canal after separation.
Thermal Loss	Not shown	Heat (Thermal Loss)	Represents energy losses generated during the separation process.
Storage Function	Simple storage block	Store Collected Algae-Debris	Clarifies that both algae and debris are temporarily stored before further processing.

Feature	Initial Diagram	Current Diagram	Reason for Change
Storage Monitoring	Not available	Monitor Storage Level	Allows continuous monitoring of the storage tank to prevent overflow.
Tank Full Signal	Not included	Alarm / Stop Collection and Signal: Tank Full	Automatically stops the collection process when the storage tank reaches its maximum capacity.
Debris Separation	Not included	Separation of Algae from Debris	Separates useful algae from unwanted waste materials before final disposal. Because this functional sketch is AI-generated, it serves as an illustrative abstraction and does not fully represent the finalized mechanical architecture or intended technical details of the conceptual storage or disposal.
Waste Handling	Not included	Remove Waste & Debris	Introduces a dedicated process for removing non-organic waste such as plastics and stones.
Waste Output	Not specified	Waste/Debris	Clearly defines the waste stream leaving the system after separation.
Energy Output	Not represented	Energy Loss (Heat)	Completes the system's energy flow by identifying energy losses.
Signal Flow	Limited control signals	Start Signal, System Positioned Signal, and Tank Full Signal	Improves system automation and provides better operational control.
Material Flow	Simple material flow	Separate Algae, Water, and Debris flows	Makes the functional model easier to understand by distinguishing the different material streams.

2. Morphological Box Synthesis

The Morphological Box decomposes the AlgaNova system into nine technical sub-functions to synthesize three distinct concept variants across a spectrum of automation:

- **Concept 1 (Manual Base):** Relies on manual paddling, visual navigation, a harvest trident, perforated basket separation, and manual dumping.
- **Concept 2 (Hybrid Power):** Combines pedal/battery propulsion with an electrically assisted conveyor, perforated belt filtration, and strain-gauge load monitoring.
- **Concept 3 (Conveyor System):** Features submerged electric propellers, sensor/camera navigation, a powered conveyor, IP67 ultrasonic volume monitoring, and an articulated rear-tipping hopper.

NOTE: All three variants converge on coarse screening/sedimentation for secondary algae debris separation.

Figure 5 is the developed morphological box:

Week 2 Deliverable

Morphological Box Project / Product													
Project: AlgaNova Group: 9 Revision: 1 Date: 22/07/2026													
ID	Function	Solution 1	Solution 2	Solution 3	Solution 4	Solution 5	Solution 6	Concept 1		Concept 2		Concept 3	
								name	Human base Collection	name	Hybrid Power collection	name	Conveyor system
								Selection n	pick one concept for each function	Selection n	pick one concept for each function	Selection	pick one concept for each function
1	Mobility	Walking	Paddeling	Swimming	Submerged Electric Propellers	Surface Paddle wheel Drives	Hybrid system (pedal battery power assist propulsion)	2	Paddeling	6	Hybrid system (pedal battery power assist propulsion)	4	Submerged Electric Propellers
2	Navigate to Area	Human eye	GPS Navigation System	AI Automated Navigation (Through Satellite)	Remote control Navigation (Semi Automated)	Embedded Motion sensors and Camera		1	Human eye	1	Human eye	5	Embedded Motion sensors and Camera
3	Collect Algae-Water-Debris Mixture		Trident	Collection Net	Vaccum water Suction system	ELEctrical power assisted Manual Conveyor system	Electrical powerd Conveyor system	Lifting scoop	1	Trident	4	ELEctrical power assisted Manual Conveyor system	5
4	Separate Algae-Debris from water	Perforated Basket	Perforated Belt Conveyor	Net				1	Perforated Basket	2	Perforated Belt Conveyor	2	Perforated Belt Conveyor
5	Store collected Algae-Debris	Human eye	Perforated Basket	Sequential Dumping spots				3	Sequential Dumping spots	2	Perforated Basket	2	Perforated Basket
6	Monitor Storage Level		Weight-Based Monitoring (Strain Gauge Load Cells)	Level/Volume Monitoring (IP67 Ultrasonic Distance Sensors)				1	Human eye	2	Weight-Based Monitoring (Strain Gauge Load Cells)	3	Level/Volume Monitoring (IP67 Ultrasonic Distance Sensors)
7	Alarm/stop Collection		Person	Mechanical Bell system	Speaker			1	Person	2	Mechanical Bell system	3	Speaker
8	Separation of Algae From Debris	Coars screening/ Sedimentation						1	Coars screening/ Sedimentation	1	Coars screening/ Sedimentation	1	Coars screening/ Sedimentation
9	Remove waste & Debris	Manual Dumping	Articulated Rear-Tipping Hopper					1	Manual Dumping	1	Manual Dumping	2	Articulated Rear-Tipping Hopper

Figure 5: Morphological Box:

3. Utility Value Analysis (*Nutzwertanalyse*)

Week 2 Deliverable

Group: 9														
Revision: 1														
Date: 7/24/2026														
Transferred data from Requirements List - No editing here!						Concept 1			Concept 2			Concept 3		
ID	Requirement	Target / Criterion	Unit	Demand / Wish	Weight	Degree of Fulfillment	value [0...10]	weighted value	Degree of Fulfillment	value [0...10]	weighted value	Degree of Fulfillment	value [0...10]	weighted value
1	Water depth operation	0 - 1	m	D	0.07	0-0.2m	2	0.14	0-0.5m	5	0.35	0-0.5m	5	0.35
2	Operating speed	~1	Km/h	D	0.05	1->6 Km/h	9	0.45	1->4 Km/h	8	0.4	1->4 Km/h	8	0.4
3	Noise emission	~60	dB	W	0.03	~5dB	10	0.3	60 -> 70dB	5	0.15	60 -> 80dB	4	0.12
4	Energy consumption	<500	W	D	0.06	0W	9	0.54	250W	8	0.48	450W	6	0.36
5	Operating Duration	4->6	h	D	0.04	3h	7	0.28	4h	8	0.32	4h	8	0.32
6	Algae storage capacity	200 -> 250	Kg	D	0.05	80Kg	2	0.1	100Kg	7	0.35	100Kg	5	0.25
7	Machine weight	<500	Kg	W	0.07	Canoe + Gear + 2Crew =210Kg	10	0.7	Machine + Crew ~450Kg	7	0.49	Machine ~450Kg	7	0.49
8	Regular Maintenance (Cleaning)	100	Euros	D	0.04	10 Euros	10	0.4	100 Euros	10	0.4	150 Euros	8	0.32
9	Harmful emissions	None	None	D	0.05	Zero Emission	10	0.5	None	10	0.5	None	10	0.5
10	Corrosion Resistance	Visual Inspection	None	W	0.03	Material mostly made of plastic	9	0.27	On a scale from 1-10: we put it at 7	7	0.21	On a scale from 1-10: we put it at 7	6	0.18
11	Compact design (Surface Area)	<5	m^2	W	0.04	~3.25	8	0.32	~4	10	0.4	4 -> 5	10	0.4
12	Attractive appearance	public opinion on design	None	W	0.03	On a scale from 1-10: we put it at a 7	7	0.21	On a scale from 1-10: we put it at a 8	8	0.24	On a scale from 1-10: we put it at a 8	8	0.24
13	Durability	10 years/warranty		D	0.03	Check	10	0.3	Check	10	0.3	Check	10	0.3
14	Machine Safety & Electrical and Radio Regulations	Class 1	None	D	0.03	Check	10	0.3	Check	10	0.3	Check	10	0.3
15	Battery Safety Regulations	UN 38.3 Certification.	None	D	0.03	check	10	0.3	Check	10	0.3	Check	10	0.3
16	Comply to the Federal Waterway Permit (Strom- und Schifffahrtpolizeiliche Genehmigung)	Conform to legal requirements	None	D	0.04	check	10	0.4	Check	10	0.4	Check	10	0.4
17	Water Use Permit (Wasserrechtliche erlaubnis)	Conform to legal requirements	None	D	0.04	check	10	0.4	Check	10	0.4	Check	10	0.4

18	Biodiversity & Habitat Protection (Naturschutzrecht)	Conform to legal requirements	None	D	0.03	Check	10	0.3	Check	10	0.3	Check	10	0.3
19	Scalability	Adaptable to Canal width	m	W	0.02	Check	10	0.2	Check	10	0.2	Check	10	0.2
20	Water proof	IPX6	None	D	0.04	Check	10	0.4	Check	10	0.4	Check	10	0.4
21	Must not block canal Navigation	Stoppage time of other simultaneously occurring activities	min	D	0.03	Check	10	0.3	Check	10	0.3	Check	10	0.3
22	Preserve canal infrastructure	No reconstruction nor destruction of the currents canal's infrastructure	None	D	0.05	Check	10	0.5	Check	10	0.5	Check	10	0.5
23	Water Resistant	Should Operable in hot,sunny,m moist and rainy weather conditions	None	D	0.02	Check	10	0.2	Check	10	0.2	Check	10	0.2
24						(20->25) Kg	3		(~50)Kg	5		(~70)Kg	6	
25				Total Weight				0			0			0
26				Check			Σ	7.81		Σ	7.89		Σ	7.53

Methodology

To determine the optimal system architecture, a weighted value analysis (utility analysis) evaluates the three synthesized concept variants against the 23 functional, physical, and regulatory requirements. Each requirement is assigned a weighting factor (t_i) totaling 1.00 (100%). Individual concepts are rated on a degree of fulfillment scale ($v_i \in [0, 10]$), yielding a weighted utility score ($w_i = t_i \cdot v_i$) per parameter.

$$\sum W_{concept1} = 7.81 \quad | \quad \sum W_{concept2} = 7.89 \quad | \quad \sum W_{concept3} = 7.53$$

Analytical Comparison

- **Concept 2 (Hybrid power collection)** Achieves the highest utility score by establishing an optimal balance between operational performance and system efficiency. It yields superior ratings in operating depth coverage (0–0.5 m, rating 5), storage capacity (100 kg, rating 7), and low maintenance costs (100 Euros, rating 10), while maintaining full compliance across all legal and safety indicators.
- **Concept 1 (Human Base Collection)** Performs strongly in energy independence (0 W, rating 9), weight (210 kg, rating 10), and low maintenance costs (10 Euros, rating 10). However, its utility is constrained by limited operational depth (0–0.2 m, rating 2) and reduced storage capacity (80 kg, rating 2).
- **Concept 3 (Conveyor System):** Delivers robust functional performance but receives penalties due to elevated power consumption (450 W, rating 6), higher noise emissions (60–80 dB, rating 4), and increased maintenance expenditure (150 Euros, rating 8).

Conclusion

Concept 2 is selected as the primary design baseline for detailed technical development based on its maximum overall utility score.

Functional Sketch Analysis: Concept 2 (Dual-Paddle Algae Collection Machine)

Figure 6 shows the functional sketch

System Architecture & Structural Layout

The design features a floating catamaran hull equipped with dual lateral paddle wheels for propulsion and fluid guidance, a central inclined perforated conveyor, an elevated twin-seat control cabin with a signaling beacon, and a high-capacity rear storage hopper.

Operational Workflow:

1. Collect: Dual paddle wheels direct surface algae and debris into the central intake.
2. Separate: The perforated conveyor lifts solid biomass while gravity-draining clean water back into the canal.
3. Store: Screened algae-debris deposits into the rear storage hopper.
4. Monitor & Alarm: Level sensors monitor fill status, triggering an audio-visual alarm and system stop upon reaching full capacity.
5. Separate & Remove: Secondary screening isolates excess moisture before the waste is offloaded onshore.

Design Notes: Because this functional sketch is AI-generated, it serves as an illustrative abstraction and does not fully represent the finalized mechanical architecture or intended technical details of the concept.

CONCEPT 2

DUAL-PADDLE ALGAE COLLECTION MACHINE

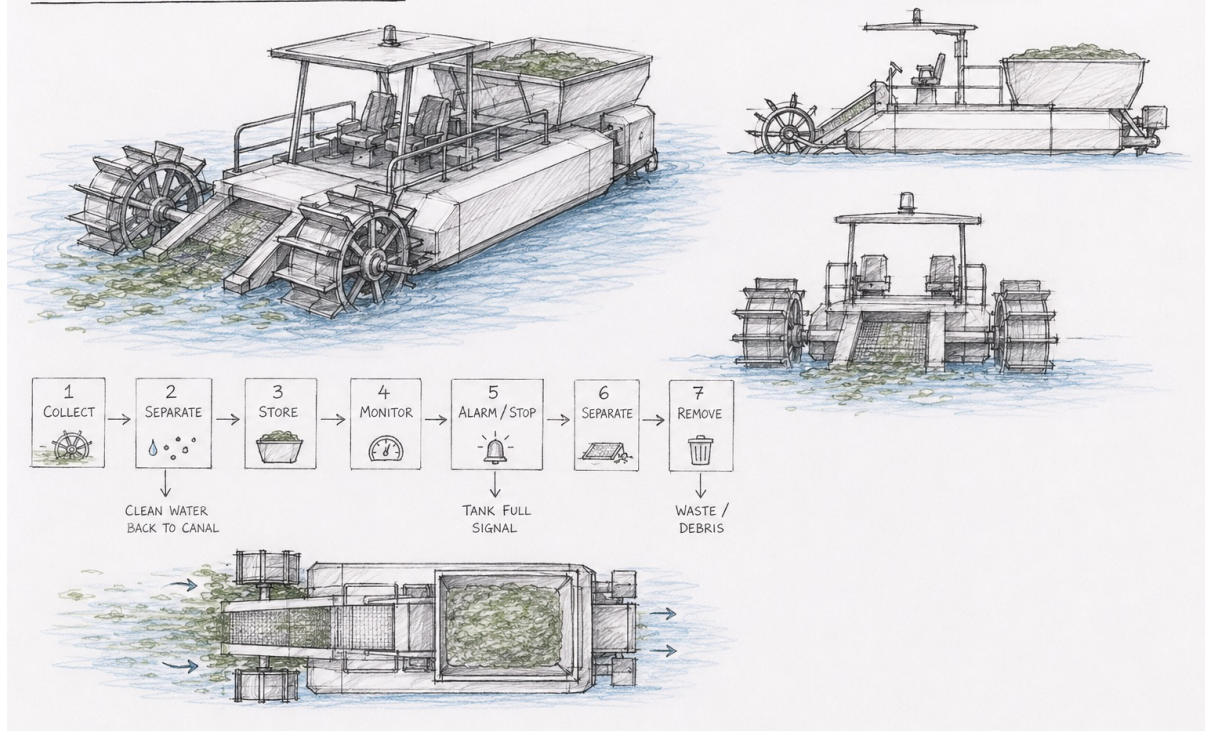


Figure 6: Functional Sketch

Conclusion & Future Outlook

The conceptual design and value analysis successfully established Concept 2 as the optimal baseline for the AlgaNova system, striking the best balance between performance, operational feasibility, and regulatory compliance.

Moving forward, several overarching challenges must be addressed during detailed development:

- **Operational & Environmental Variability:** Managing unpredictable canal conditions, such as varying water currents, floating debris types, and fluctuating algal density.
- **System Integration & Efficiency:** Optimizing the trade-offs between battery weight, drive-system power consumption, and total structural payload.
- **Durability & Regulatory Adherence:** Maintaining long-term material resilience against corrosion while meeting all local environmental and waterway navigation standards.

Future work will transition the project from conceptual evaluation to detailed mechanical engineering, component sizing, and physical prototype testing.

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